

Zixuan SU

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EDUCATION BACKGROUND

Dalhousie University, Department of Electrical and Computer Engineering 09/2025-Ongoing
Ph.D. in Electrical Engineering, supervised by **Zhizhang (David) Chen** (Fellow, IEEE; Fellow, CAE; Fellow, EIC)
City University of Hong Kong, Department of Electrical Engineering 09/2020-06/2025
Master of Science in Electronic Information Engineering with Merit.
Fuzhou University, Department of Physics and Information Engineering 09/2020-06/2024
Bachelor of Engineering in Communication Engineering, GPA: 3.32/4.00 (rank 17/74).

AWARDS

First Prize in The Chinese Mathematics Competition (CMC) (Nov. 2023)
National Encouragement Scholarship (5000 RMB, **Three times**)
Elite Student Honor at Fuzhou University Overseas Elite Training Camp (18000 RMB)
Second Prize for Academic Performance Scholarship (1000 RMB)
Third Prize in National College Student Information Technology Competition Fujian Division (Aug. 2023)
Third Prize in **National Undergraduate Electronic Design Contest (Ti Cup)** Fujian Division (Aug. 2022)
Third Prize for Academic Performance Scholarship (500 RMB, Twice)
Spiritual Civilization Scholarship (400 RMB)

PUBLICATIONS

J1. Ziheng Zhou, Zubing Chen, **Zixuan Su**, Min Li, Zhechen Zhang, Yuehe Ge, Yueming Gao and Zhizhang (David) Chen, "Pixel-Metastructure-Assisted Subarray Decoupling for Compact MIMO Antenna Designs", **IEEE Transactions on Antennas and Propagation (IF: 5.8, JCR Q1)**, May. 2025. (Published)
J2. Ziheng Zhou, **Zixuan Su**, Yanfeng Zhou, Zhechen Zhang, Zhimeng Xu and Zhizhang (David) Chen, "Antenna Matching Network Topology Optimization: Theoretical Modeling and Numerical Experiments", **Experimental Technology and Management (a core Chinese Journal)**, Aug. 2024. (Published)
C. Ziheng Zhou, **Zixuan Su**, and Zhizhang (David) Chen, "Waveguide-emulated index-near-zero media for antenna applications", **2024 International Conference on Microwave and Millimeter Wave Technology (ICMMT)**, Beijing, China, May. 2024. (Invited talk)

PROJECT EXPERIENCE

Surrogate Model Accelerated Rectangular Dielectric Resonator Antenna Design 03/2025-05/2025
•Tutor: Kwok Wa Leung (IEEE Fellow)
•Utilized Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) for optimizing a RDRA's volume at target resonant frequency.
•Leveraged Backpropagation Neural Networks (BPNN) as surrogate models to approximate complex electromagnetic relationships between geometric parameters and resonant frequencies.
Single-Stage Amplifier Design 02/2025-05/2025
•**Group Leader** (Supervised by Prof. David Ng, Adjunct Professor, The University of Hong Kong)
•Implemented a current mirror circuit to optimize amplifier balance and overall performance.
•Recognized as **Outstanding Group** in the final oral presentation.
Design of Signal Modulation Measurement Device based on FPGA 05/2022-08/2022
•Leveraged Fast Fourier Transform (FFT) to perform modulation characteristic analysis of AM and FM signals in the frequency domain.
•Utilized MATLAB to visually track signal modulation and made a graphical interface by STM32 for results, making it easier for users to grasp modulation details.

EXTRACURRICULAR ACTIVITIES

Fuzhou University library Multimedia Operations Department 09/2020-06/2024
•**Position hold:** *Section Leader*
•Led the group with responsibilities such as poster design, event shooting and publicity work.
•**Awards:** One of our projects was selected as the **2021 National Reading Promotion Demonstration Project** by the Chinese Library Association.
Teaching Assistant, University Physics 06/2023-12/2024
•Evaluated and provided detailed feedback on over 500 assignments for 70+ students, ensuring fair and consistent grading standards.
•Conducted weekly tutorial sessions and office hours to reinforce lecture materials and assist students with problem-solving.
•**Awards:** Recognized by the supervising professor for **Excellent TAs** in the final assessment.
HarmonyOS Development Courses 08/2022-08/2023
•Received training of this distributed operating system, gained the comprehensive understanding, technical features and principles for application development based on HarmonyOS system.
•**Awards:** Skill Certificate of "Internet of Things" by Huawei Technologies Co., Ltd.

OTHERS

•Programming language: MATLAB, C, Verilog HDL
•EDA & Simulation Tools: SPICE, HFSS
•Hobbies: Photoshop, Premiere, Handwriting, Badminton